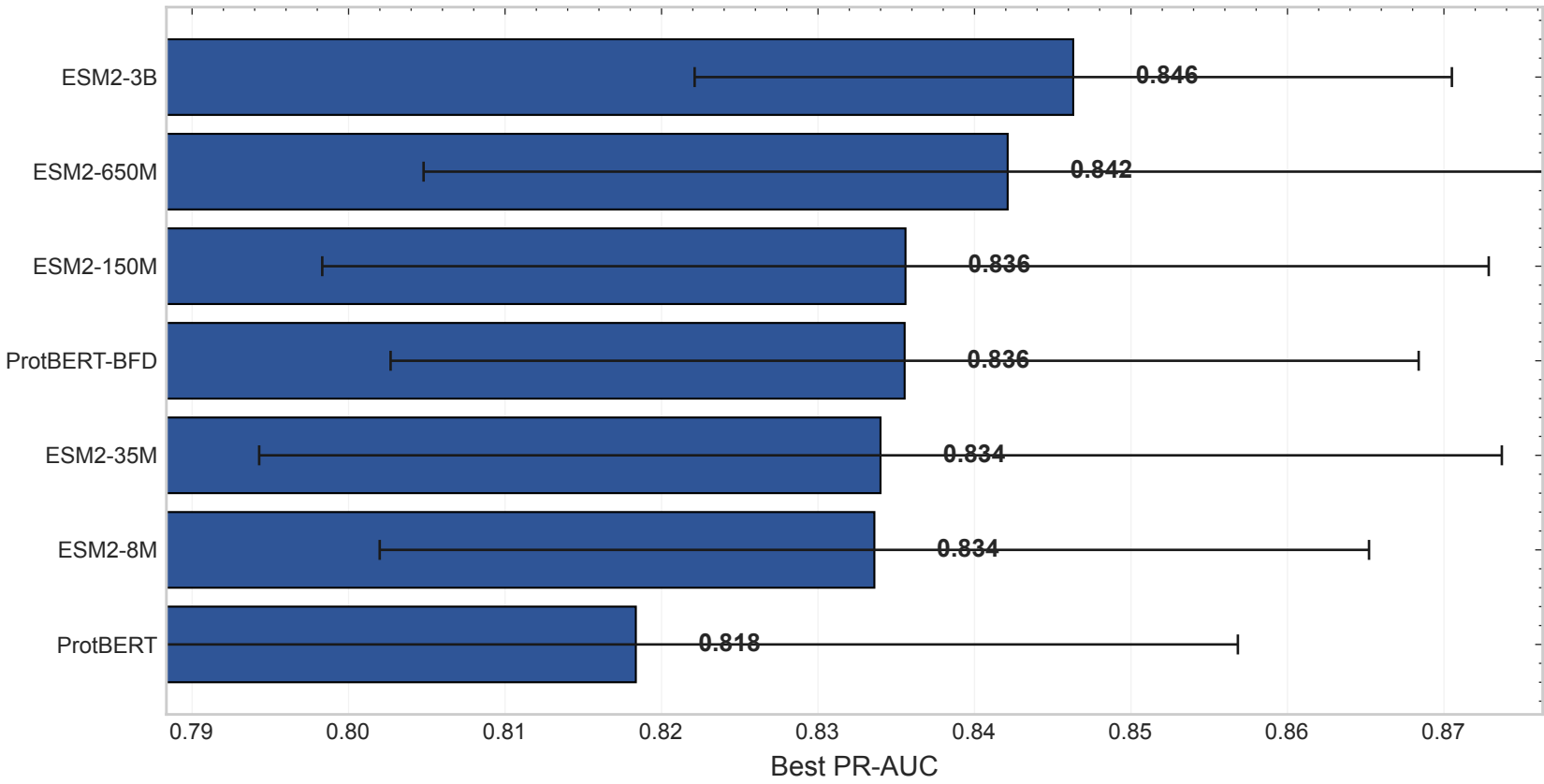
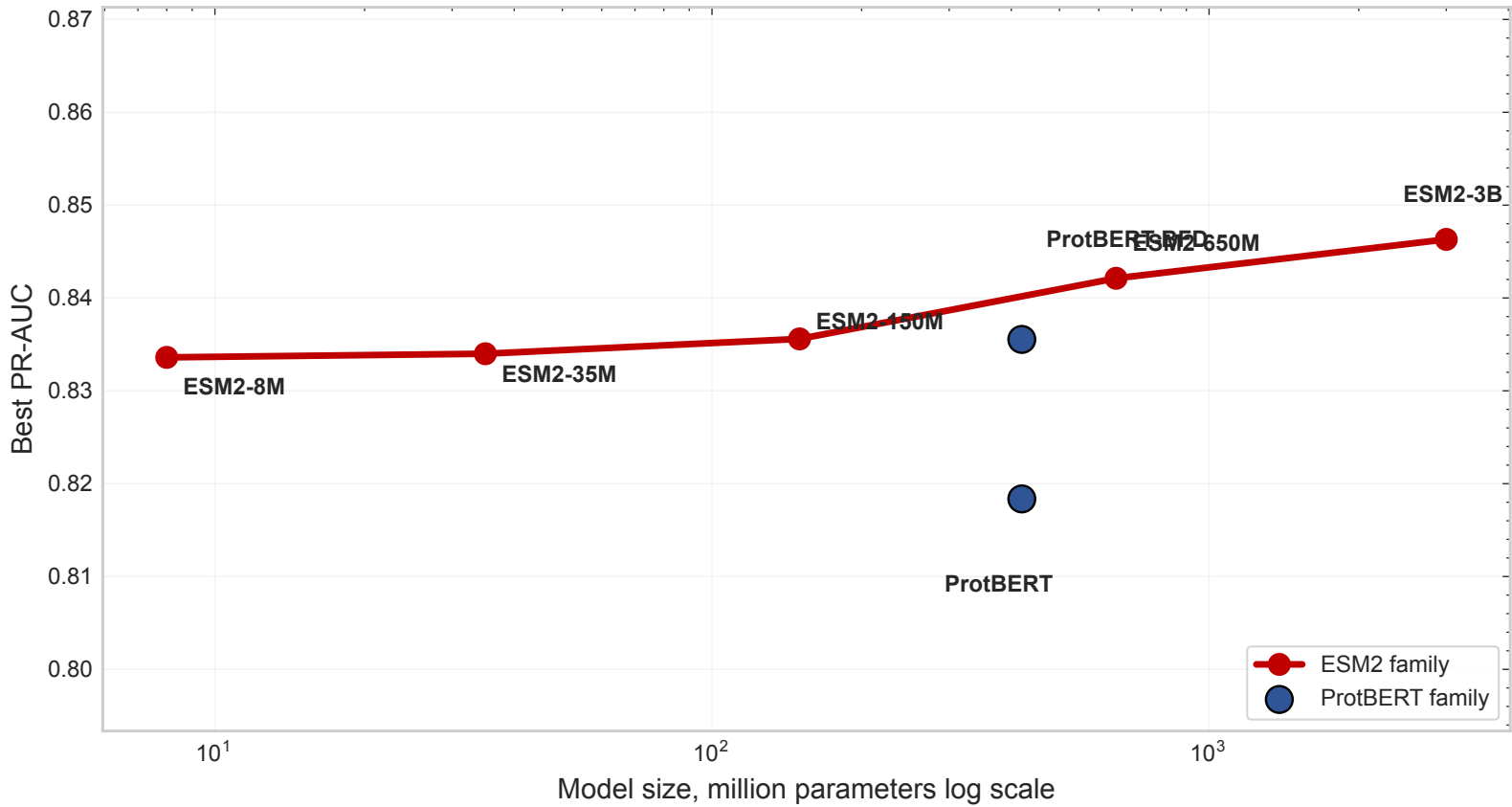


Figure 2. Benchmarking protein language models for E3/DUB specificity prediction

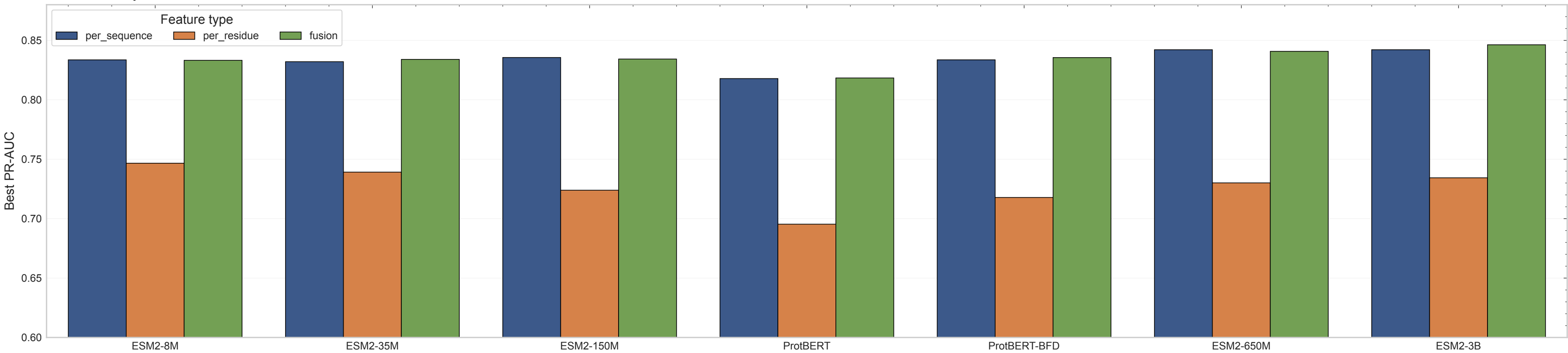
A. Best performance by protein language model



B. Parameter scaling and performance saturation



C. Feature representation benchmark



D. Best configuration per embedding model

Model	Params M	Best feature	Algorithm	ROC-AUC	PR-AUC	Brier
ESM2-3B	3000	fusion	xgboost	0.759	0.846	0.215
ESM2-650M	650	per_sequence	xgboost	0.758	0.842	0.215
ESM2-150M	150	per_sequence	xgboost	0.754	0.836	0.217
ProtBERT-BFD	420	fusion	xgboost	0.751	0.836	0.218
ESM2-35M	35	fusion	xgboost	0.754	0.834	0.214
ESM2-8M	8	per_sequence	xgboost	0.748	0.834	0.222
ProtBERT	420	fusion	xgboost	0.728	0.818	0.228